The successor function is lambda definable

The Open Logic Project contributors

openlogic:lambda-calculus/lambda-definability:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/olp-lambda-calculus-lambda-definability/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Card cut from the FREE slice of the lit pool for unit olp-lambda-calculus-lambda-definability; no cross-unit references occur inside the span.

Editorial scope. Single proposition card cut from the Open Logic Project unit olp-lambda-calculus-lambda-definability (pinned-original-source.tex); statement and proof are the author's own text at source lines 415-418 and 420-438.

Allowed standard prerequisites (editorial).

- Church numeral encoding (lambda-definability unit, earlier section: the numeral for n is the term lambda f x. f^n x). restated inside the proof at source line 433. The reduction n f x -> f^n x at proof lines 432-434 is the engine of the verification.

Proposition 0.1. *The successor function* S *is lambda definable.*

Proof. A term that lambda defines the successor function is

$$\text{Succ} \equiv \lambda a.\, \lambda f x.\, f(a f x).$$

Given our conventions, this is short for

$$\text{Succ} \equiv \lambda a.\, \lambda f.\, \lambda x.\, (f((a f) x)).$$

Succ is a function that accepts as argument a number a, and evaluates to another function, $\lambda f x.\, f(a f x)$. That function is not itself a Church numeral. However, if the argument a is a Church numeral, it reduces to one. Consider:

$$(\lambda a.\, \lambda f x.\, f(a f x))\, \overline{n} \to \lambda f x.\, f(\overline{n} f x).$$

The embedded term $\overline{n} f x$ is a redex, since $\overline{n}$ is $\lambda f x.\, f^n x$. So $\overline{n} f x \to f^n x$ and so, for the entire term we have

$$\text{Succ}\ \overline{n} \twoheadrightarrow \lambda f x.\, f(f^n(x)),$$

i.e., $\overline{n+1}$. □